

# Recipe Pipeline

## Code:

```
from airflow.decorators import dag, task

from datetime import datetime
import random

@dag(
    dag_id="recipe_pipeline",
    schedule="@daily",
    catchup=False,
    tags=["recipe"]
)
def recipe_pipeline():

    @task
    def get_ingredients(ds=None) -> dict:
        print(f"Getting ingredients for {ds}")
        return {
            "dish": "Pasta",
            "ingredients": ["pasta", "tomato sauce", "cheese", "garlic"]
        }

    @task
    def prepare_dish(recipe: dict) -> dict:
        dish = recipe["dish"]
        ingredients = recipe["ingredients"]

        print(f"Preparing {dish}...")
        for i, item in enumerate(ingredients, start=1):
            print(f"{i}. {item}")

        return {
            "dish": dish,
            "ingredient_count": len(ingredients),
            "status": "ready"
        }

    @task
    def serve_dish(result: dict) -> str:
        print(f"Serving {result['dish']} - {result['ingredient_count']} ingredients - Status: {result['status']}")
        return f"Served: {result['dish']}"

    @task.branch
    def check_rating(result: dict) -> str:
```

```

rating = random.randint(1, 10)
print(f"Customer rating: {rating}/10")

if rating >= 7:
    return "positive_review"
else:
    return "needs_improvement"

@task
def positive_review() -> None:
    print("Great dish! Marking as customer favourite.")

@task
def needs_improvement() -> None:
    print("Recipe needs work. Adding to the improvement list.")

@task(trigger_rule="none_failed")
def end_of_service(served: str, ds=None) -> None:
    print(served)
    print(f"Kitchen closed for {ds}. All done!")

ingredients = get_ingredients()
prepared = prepare_dish(ingredients)
served = serve_dish(prepared)

review = check_rating(prepared)

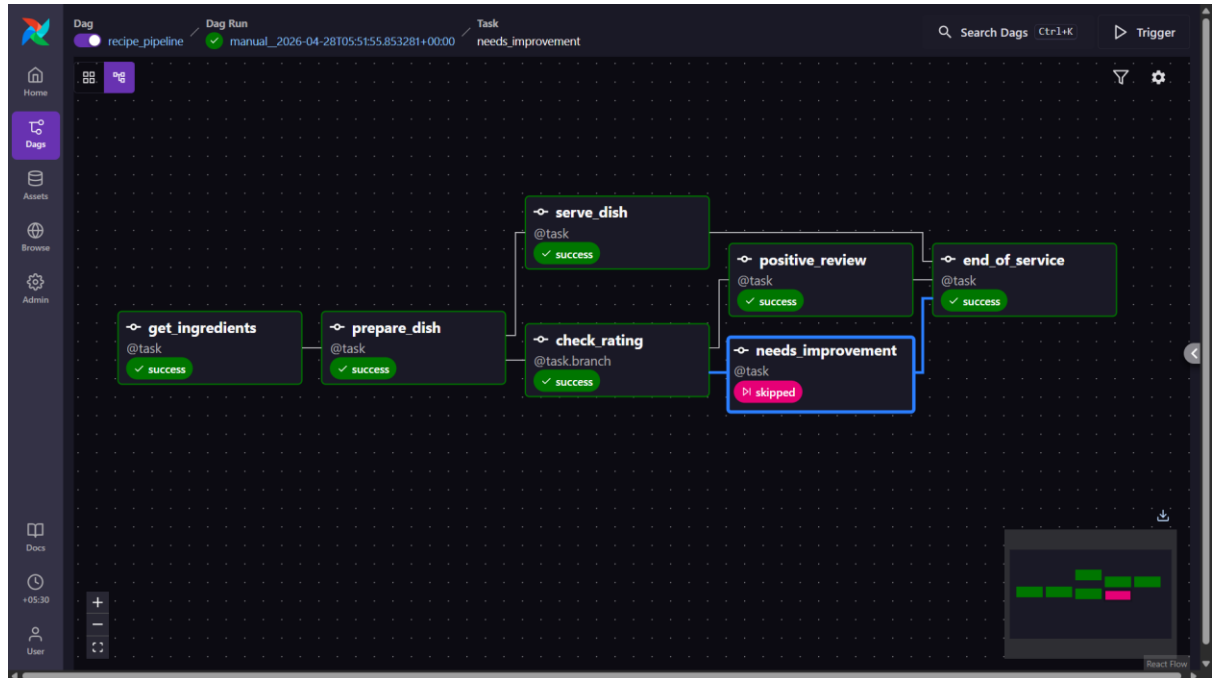
review >> [positive_review(), needs_improvement()]

end_of_service(served)

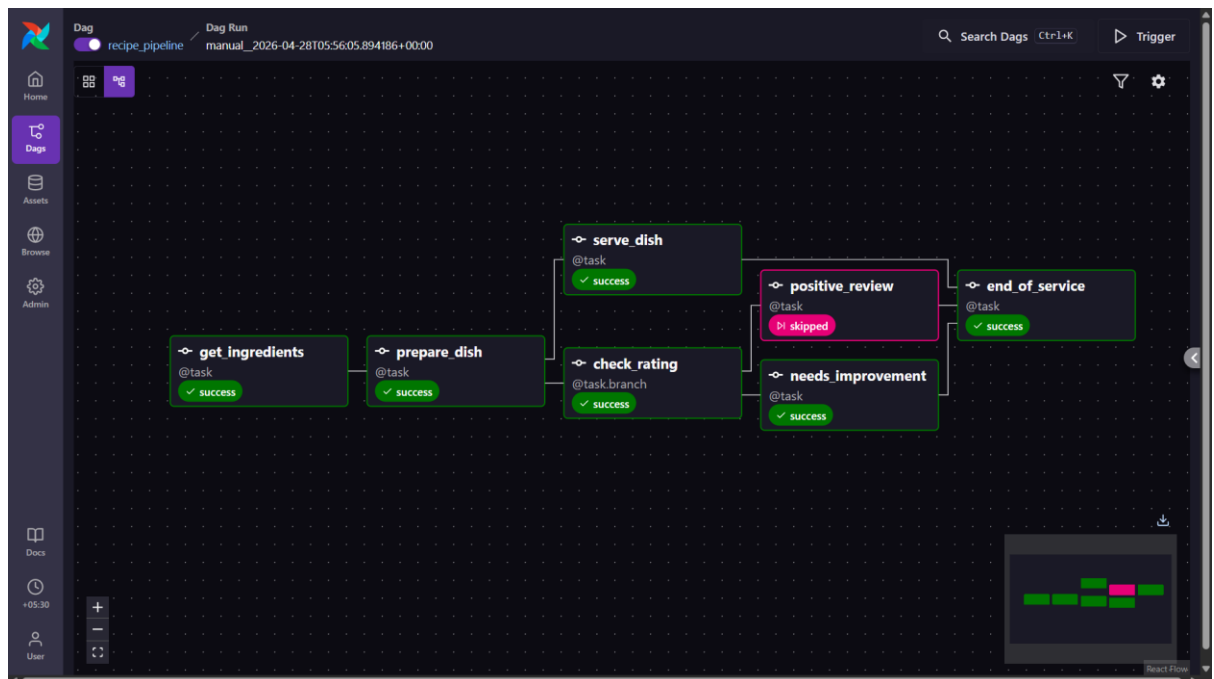
dag_run = recipe_pipeline()

```

1. Graph View — Run A: positive\_review GREEN, needs\_improvement pink (skipped), end\_of\_service GREEN



2. Graph View — Run B: needs\_improvement GREEN, positive\_review pink (skipped), end\_of\_service GREEN



### 3. Task Log of prepare\_dish — showing the dish name and numbered ingredient list printed

The screenshot displays the Airflow web interface. On the left, a DAG named 'recipe\_pipeline' is shown with a task 'prepare\_dish' highlighted. The main panel shows the task log for 'prepare\_dish', which is in a 'Success' state. The log details include the operator '@task', start date '2026-04-28 11:27:51', end date '2026-04-28 11:27:52', duration '00:00:00.227', and DAG version 'v2'. The log messages show the task preparing pasta and listing ingredients: 1. pasta, 2. tomato sauce, 3. cheese, and 4. garlic. The log is highlighted with a yellow box.

```
Log message source details sources=["/usr/local/airflow/logs/dag_id-recipe_pipeline/run_id-manual_2026-04-28T05:57:50.104512+00:00/"]
2 [2026-04-28 11:27:51] WARNING - Astro managed secrets backend is disabled
3 [2026-04-28 11:27:51] INFO - DAG bundles loaded: dags-folder
4 [2026-04-28 11:27:51] INFO - Filling up the DagBag from /usr/local/airflow/dags/Assignment-2/recipe_dag.py
5 [2026-04-28 11:27:51] WARNING - The 'airflow.decorators.dag' attribute is deprecated. Please use 'airflow.sdk.dag'. category=
6 [2026-04-28 11:27:51] WARNING - The 'airflow.decorators.task' attribute is deprecated. Please use 'airflow.sdk.task'. category=
7 [2026-04-28 11:27:51] INFO - TaskInstance Details: dag_id=recipe_pipeline task_id=prepare_dish dagrun_id=manual_2026-04-28T05:57:50.104512+00:00
8 [2026-04-28 11:27:51] INFO - Preparing Pasta...
9 [2026-04-28 11:27:51] INFO - 1. pasta
10 [2026-04-28 11:27:51] INFO - 2. tomato sauce
11 [2026-04-28 11:27:51] INFO - Done. Returned value was: {'dish': 'Pasta', 'ingredient_count': 4, 'status': 'ready'}
12 [2026-04-28 11:27:51] INFO - Pushing xcom ti=RuntimeTaskInstance(id=UUID('019dd2aa-4faa-7cd8-9717-93ab1a92a539'), task_id='prepare_dish', dag_id='recipe_pipeline', run_id='manual_2026-04-28T05:57:50.104512+00:00')
13 [2026-04-28 11:27:51] INFO - 3. cheese
14 [2026-04-28 11:27:51] INFO - 4. garlic
```

### 4. Task Log of end\_of\_service — showing the served string and the closing message with ds

The screenshot displays the Airflow web interface. On the left, a DAG named 'recipe\_pipeline' is shown with a task 'end\_of\_service' highlighted. The main panel shows the task log for 'end\_of\_service', which is in a 'Success' state. The log details include the operator '@task', start date '2026-04-28 11:27:55', end date '2026-04-28 11:27:55', duration '00:00:00.105', and DAG version 'v2'. The log messages show the task serving pasta and closing the kitchen. The log is highlighted with a yellow box.

```
Log message source details sources=["/usr/local/airflow/logs/dag_id-recipe_pipeline/run_id-manual_2026-04-28T05:57:50.104512+00:00/"]
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6 [2026-04-28 11:27:55] WARNING - The 'airflow.decorators.task' attribute is deprecated. Please use 'airflow.sdk.task'. category=
7 [2026-04-28 11:27:55] INFO - TaskInstance Details: dag_id=recipe_pipeline task_id=end_of_service dagrun_id=manual_2026-04-28T05:57:50.104512+00:00
8 [2026-04-28 11:27:55] INFO - Done. Returned value was: None
9 [2026-04-28 11:27:55] INFO - Served: Pasta
10 [2026-04-28 11:27:55] INFO - Kitchen closed for 2026-04-28. All done!
```